

start: a first guess $\hat{x}_0$ with variance p_0

predict (motion)

$$\bar{x} = \hat{x} + u$$

$$\bar{p} = p + Q$$

the bell moves and widens

$\bar{x}, \bar{p}$

update (reading z)

$$K = \bar{p} / (\bar{p} + R)$$

$$\hat{x} = \bar{x} + K(z - \bar{x})$$

$$p = (1 - K) \bar{p}$$

the bell moves towards z and narrows

$\hat{x}, p$: next step